

E-commerce Order Processor

Scope: implemented order processing, VIP discounts, and inventory rejection behavior

BUSINESS REQUIREMENTS

REQ-001 **Standard Processing**

The processor shall calculate an order subtotal by multiplying each item quantity by its unit price. For valid standard-customer orders with sufficient inventory, it shall reduce inventory by the ordered quantities and return the subtotal as the order total, with no discount.

REQ-002 **VIP Discounts**

The processor shall apply a 10% discount to the subtotal for VIP customers. Customer types other than VIP shall receive no discount. The order result shall report the subtotal, discount, and discounted total.

REQ-003 **Error Handling for Negative Inventory**

The processor shall reject orders when inventory is negative or insufficient for the requested quantities. A rejected order shall leave inventory unchanged; inventory validation shall complete before any stock is reduced.
